Dataset Details • Specification Sheet

CARE_To_Compare Benchmark • SCADA Telemetry • Condition Monitoring Reference

OPEN SCIENCE BENCHMARK

1. DATASET SPECIFICATION & DOWNLOAD HYPERLINK

DIRECT ZENODO REPOSITORY HYPERLINK (CLICK TO ACCESS)

https://zenodo.org/records/15846963 — Open on Zenodo →

BENCHMARK TITLE	ZENODO RECORD ID	DATA VOLUME	TEMPORAL SPAN
CARE_To_Compare	15846963	5.5 GB (102M Points)	89 Turbine-Years

2. MULTI-FARM ARCHITECTURE & TELEMETRY COVERAGE

WIND FARM	ENVIRONMENT	TURBINES	DATASETS (EVENTS)	SENSOR CHANNELS
Wind Farm A	Onshore (Portugal)	5 (IDs: 0, 10, 11, 13, 21)	22 (12 Fault, 10 Normal)	86 channels (EDP Open Data)
Wind Farm B	Offshore (Germany)	Fleet-scale	15 (6 Fault, 9 Normal)	257 channels (Marine SCADA)
Wind Farm C	Offshore (Germany)	Fleet-scale	58 (27 Fault, 31 Normal)	957 channels (Deep telemetry)
Combined Fleet	**Onshore & Offshore**	**36 Turbines**	**95 Datasets (44 Fault, 51 Normal)**	**~1,300 unique SCADA signals**

3. STRUCTURE OF MONITORED EVENTS (STANDARD IEC 61400-25)

EVENT WINDOW	OPERATIONAL DESCRIPTION
Historical Baseline	**1 full year** (~52,560 rows) of healthy normal operation establishing seasonal thermal baselines.
Evaluation Window	**2 to 4 weeks** (336 to 672 hours) immediately preceding breakdown or continued normal running.
Sampling Rate	**10-minute continuous intervals** (IEC 61400-25: Mean, Min, Max, Std Dev for each signal).

DATA ATTRIBUTE	FORMAT & PROTOCOL
File Format	Semicolon-delimited CSV (`;`), UTF-8 encoding.
Timestamps	ISO 8601 UTC (`YYYY-MM-DD HH:MM:SS`).
Class Balance	44 Anomaly Datasets vs. 51 Healthy Normal Controls.

4. GROUND-TRUTH FAILURE MODES & KEY SCADA CHANNELS

COMPONENT	PRIMARY SENSOR CHANNELS	PHYSICAL FAILURE SIGNATURE
Gearbox	`temp_gearbox_bearing` , `temp_gearbox_oil` , `gear_ratio`	High-speed shaft bearing friction, thermal runaway, and gear slip.
Generator Bearing	`temp_gen_bearing_de` , `temp_gen_bearing_nde` , `gen_rpm`	Drive-End bearing spalling, severe localized friction up to 94°C.
HV Transformer	`temp_transformer` , `temp_ambient` , Phase L1/L2/L3	Winding insulation breakdown and severe internal ohmic heating.
Hydraulic Group	`temp_hydraulic_oil` , hydraulic pressure, blade pitch	Proportional valve pressure drop, actuator loss, and blade drift.

5. DATA HYGIENE, LEAKAGE PREVENTION & PHYSICS BOUNDARIES

Rule 1: Post-Trip Status Code Purge (No Cheating)
Raw logs contain `status_type_id` (0=Normal, 4=Downtime). Models that include status codes achieve 99.9% artificial accuracy because downtime is recorded *after* trip. Status codes were strictly purged to enforce genuine pre-trip early detection.

Rule 2: Zero Synthetic Data (Veto on SMOTE)
Synthetic oversampling creates physically impossible combinations (e.g. 2,000 kW power at 0 m/s wind). The dataset strictly retains 100% genuine sensor telemetry, preserving thermodynamic and Betz aerodynamic validity.